

# “PAPER\_{0:D3}” -f [int]# PAPER #120 — UQFF Astronomical Systems Catalog: 24-System Parameter Reference

**Title:** Complete Catalog of Astronomical Systems Used in UQFF Calculations — Parameters, Verification Sources, and Equation Assignments

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**Framework:** UQFF Star-Magic ( $\kappa = 0.0005/\text{day}$ ,  $[SSq] = 0.57$ ,  $\beta_i = 0.61$ )

**Date:** March 2026

**Source:** Grok thread 2fe4fa3e — DeepSearch extraction of “UQFF Equations Across Astrophysical Systems\_22Sept2025.docx” (393 pages)

**Index Slot:** §1.16 UQFF Equation Systems Reference

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

## Abstract

**Key UQFF calibrated constants:**  $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$ ;  $[SSq] = 5.7\text{e-}1$ ;  $H_{SCm} \approx 9.9\text{e-}1$ ;  $U_{UA} \approx 1.0\text{e-}4$ ;  $k_\eta = 1.0\text{e-}113$ ;  $\beta_i \approx 6.0\text{e-}1$ ;  $G = 6.674\text{e-}11 \text{ N}\cdot\text{m}^2/\text{kg}^2$

The UQFF framework has been validated and applied across 24 unique astrophysical systems spanning stellar, galactic, nuclear, particle, and cosmological domains. This paper provides a complete parameter catalog for all 24 systems, organized by type, with UQFF equation assignments, calibrated parameter values, verification data sources, and cross-references to existing whitepapers. The catalog is extracted from the 393-page “UQFF Equations Across Astrophysical Systems” corpus (verified Sept 22, 2025) as the authoritative single-document reference for system parameters used in any UQFF calculation.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. Stellar and Solar Systems

### 1.1 Sun (Solar System)

**Role in UQFF:** Heliosphere modeling via Ug2 bubble, solar wind calibration.

| Parameter                         | Value                                             | Source                    |
|-----------------------------------|---------------------------------------------------|---------------------------|
| $M_s$                             | $1.989 \times 10^{30} \text{ kg}$                 | IAU 2015                  |
| $R_b$ (heliosphere bubble radius) | $1.496 \times 10^{13} \text{ m} = 100 \text{ AU}$ | Voyager 1 crossing        |
| $\omega_s$ (rotation rate)        | $2.5 \times 10^{-6} \text{ rad/s}$                | Solar sidereal rotation   |
| $B_s$ (surface magnetic field)    | $0.0001\text{--}0.4 \text{ T}$                    | GOES 2025                 |
| $\delta_{sw}$                     | $0.01 = [SSq]/57$                                 | EP-07, PSP CDAWeb E01-E17 |

| Parameter                   | Value                                         | Source                 |
|-----------------------------|-----------------------------------------------|------------------------|
| $v_{sw}$                    | $5 \times 10^5$ m/s                           | EP-07 PSP baseline     |
| $\rho_{sw}$                 | $\approx 8 \times 10^{-21}$ kg/m <sup>3</sup> | Parker Solar Probe     |
| $H_{SCm}$                   | $\approx 1$                                   | Heliosphere SCm factor |
| $T_s$ (surface temperature) | 5778 K                                        | IAU                    |

**UQFF equations:**  $U_{g2}$  (bubble),  $U_m$  (disk strings),  $U_{b,i}$  (expansion)

**Verification:** EP-07 (PAPER\_114), Parker Solar Probe CDAWeb 2025

**Q\_wave contribution:**  $Q_{wave,\odot}$  included in 47-system mean  $3.97 \times 10^4$  J/m<sup>3</sup>

## 1.2 Westerlund 2 (Star Cluster)

**Role in UQFF:** Triadic master for turbulent outflow modeling.

| Parameter  | Value                                                    | Source             |
|------------|----------------------------------------------------------|--------------------|
| Distance   | $\approx 8$ kpc                                          | Gaia DR3           |
| Mass       | $\sim 10^4$ M?                                           | JWST 2025          |
| Age        | $\sim 2$ Myr                                             | Membership surveys |
| SFR        | High (active star formation)                             | JWST 2024          |
| $Q_{wave}$ | $\approx 3.97 \times 10^4$ J/m <sup>3</sup> (mean stack) | 47-system array    |

**UQFF equations:**  $F_{U,Triadic}$  with  $n = 13$  (plasma),  $U_m$  (turbulence)

**Verification:** JWST 2025 images, 47-system  $Q_{wave}$  stats (Jarque-Bera = 8.78,  $p = 0.012$ )

**Cross-reference:**  $Q_{wave\_47}$  statistical system

## 2. Black Hole and Galactic Center Systems

### 2.1 Sagittarius A\* (Sgr A\*)

**Role in UQFF:** SMBH anchor for  $U_{g4}$  (BH-star interaction), galactic distance  $d_g$ .

| Parameter    | Value                                              | Uncertainty  | Source                  |
|--------------|----------------------------------------------------|--------------|-------------------------|
| $M_{bh}$     | $8.15 \times 10^{36}$ kg<br>$= 4.1 \times 10^6$ M? | $\pm 4.68\%$ | Gaia DR4 2025           |
| $d_g$        | $2.44 \times 10^{20}$ m<br>$= 25,800$ ly           | $\pm 4.51\%$ | Gaia DR4 (PAPER_110)    |
| $d_g$ (UQFF) | $2.55 \times 10^{20}$ m<br>$= 27,000$ ly           | —            | UQFF calibrated         |
| $\omega_g$   | $7.3 \times 10^{-16}$ rad/s                        | —            | Galactic rotation curve |
| $U_{g4}$     | $1.8937 \times 10^{-23}$ J/m <sup>3</sup>          | —            | PAPER_110 calculated    |

**UQFF equations:**  $U_{g4}$ ,  $E_{react}$  (from BH-scale reactivity)

**Verification:** EP-06 (PAPER\_110), Gaia DR3/DR4 2025

**Note:** The UQFF uses  $d_g = 2.55 \times 10^{20}$  m (Galactic longitude model) vs Gaia's  $2.44 \times 10^{20}$  m (4.5% difference within observational uncertainty).

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## 2.2 G359.13142-0.20005

**Role in UQFF:** High-z object for shear map analysis and triadic analogs.

| Parameter            | Value                                  | Source         |
|----------------------|----------------------------------------|----------------|
| Galactic coordinates | $l = 359.13^\circ$ , $b = -0.20^\circ$ | NISP survey    |
| $\delta_\tau$        | $\approx 0.05$                         | JWST NISP 2025 |

**UQFF equations:**  $F_{U, \text{Triadic}}$ , shear maps,  $\delta_\tau$  calibration

**Verification:** JWST 2025 data, NISP catalog

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## 3. Quasar and Blazar Systems

### 3.1 3C 273 (Quasar)

**Role in UQFF:** Asymmetric jet verification of  $t_n$  reversals.

| Parameter            | Value                                    | Source                |
|----------------------|------------------------------------------|-----------------------|
| Redshift $z$         | 0.158                                    | Schmidt 1963          |
| Luminosity $L$       | $\sim 10^{46}$ erg/s = $10^{39}$ W       | MNRAS 2025            |
| Jet length           | $\sim 200$ kpc (apparent), 65 kpc proper | VLBI                  |
| Brightness asymmetry | $> 100 : 1$                              | MNRAS 482, 743 (2019) |
| $N$ (reversal count) | 13 sequential $t_n$ reversals            | PAPER_115             |
| Asymmetry ratio $R$  | $130 > 100$ target                       | EP-09 calculation     |

**UQFF equations:**  $U_{b,i}$  with  $\cos(\pi t_n)$ ,  $U_m$ ; ratio  $R = |\cos(\pi t_{n1})/\cos(\pi t_{n2})|^N$

**Verification:** EP-09 (PAPER\_115), MNRAS 482/743 2019, arXiv:2412.18250

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### 3.2 RACS J0320-35 (Quasar)

**Role in UQFF:** Navier-Stokes jet fluid verification of  $U_{b,i}$  asymmetry.

| Parameter                   | Value               | Source            |
|-----------------------------|---------------------|-------------------|
| Jet asymmetry ratio $R$     | $\approx 1.5$       | Chandra 2025      |
| $\tau_{dissip}$             | 9 Gyr               | EP-01 calculation |
| $\cos(\omega t_n)$ reversal | Sign flip confirmed | EP-01             |

**UQFF equations:**  $U_{b,i}$ , Navier-Stokes fluid in jets

**Verification:** EP-01 (PAPER\_111), Chandra X-ray Observatory 2025

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### 3.3 Blazars (Fermi LAT 4LAC)

**Role in UQFF:**  $E_{react}$  decay rate  $\kappa$  calibration across 3,743 sources.

| Parameter            | Value                                             | Source          |
|----------------------|---------------------------------------------------|-----------------|
| Sample size          | 3,743 sources (4LAC-DR3)                          | Fermi HEASARC   |
| Luminosity range     | $10^{39}$ - $10^{47}$ W                           | 4LAC-DR3        |
| $\kappa$ (measured)  | $0.000497 \pm 5\%$ day <sup>-1</sup>              | EP-05 PAPER_113 |
| 8-bin $\kappa$ error | All $< 5\%$                                       | EP-05           |
| $E_{react}$ range    | $10^{39}$ - $10^{47}$ W/m <sup>3</sup> (8 L bins) | EP-05           |

**UQFF equations:**  $E_{react} = 10^{46} e^{-\kappa t}$ ,  $L \propto E_{react}$  by blazar class

**Verification:** EP-05 (PAPER\_113), Fermi LAT 4LAC-DR4, HEASARC 2025

## 4. Galaxy Cluster and Radio Systems

### 4.1 PLCK G287.0+32.9 (Galaxy Cluster)

**Role in UQFF:** Double radio relic calibration for triadic systems.

| Parameter   | Value                            | Source      |
|-------------|----------------------------------|-------------|
| Type        | Galaxy cluster (double relic)    | Planck PSZ2 |
| $M_{500,X}$ | Calibrated to triadic mass scale | PSZ2        |
| Q_wave      | In 47-system array               | Grok corpus |

**UQFF equations:**  $F_{U,Triadic}$ ,  $M_{500,X}$  calibration

**Verification:** Planck/PSZ2 catalog

### 4.2 ASKAP J1832-0911 (Radio Transient)

**Role in UQFF:** Transient source for Q\_wave array updates and triadic analogs.

| Parameter      | Value              | Source      |
|----------------|--------------------|-------------|
| Survey         | ASKAP 2025         | ASKAP       |
| Classification | Radio transient    | ASKAP       |
| Q_wave         | In 47-system array | Grok corpus |

**UQFF equations:**  $F_{U,Triadic}$  transient variant

**Verification:** ASKAP 2025 survey

### 4.3 PSZ2 G181.06+48.47 (Galaxy Cluster)

**Role in UQFF:** Low-mass merger for double relic analogs.

| Parameter   | Value                            | Source      |
|-------------|----------------------------------|-------------|
| $M_{500,X}$ | $2.57 \times 10^{14} \text{ M?}$ | Planck 2025 |
| Merger type | Low-mass merger                  | Planck      |
| Q_wave      | In 47-system array               | Grok corpus |

**UQFF equations:**  $F_{U,\text{Triadic}}$ , merger relic analog

**Verification:** Planck 2025, PSZ2 catalog

## 5. Transient and Merger Systems

### 5.1 GW170817 (BNS Merger)

**Role in UQFF:**  $\beta_i$  calibration via r-process ejecta fraction.

| Parameter                    | Value                                       | Source            |
|------------------------------|---------------------------------------------|-------------------|
| Event date                   | August 17, 2017                             | LIGO/Virgo        |
| Total merger mass            | $\approx 2.8 \text{ M?}$                    | LIGO 2017         |
| Ejecta mass $M_{ej}$         | $\approx 0.05 \text{ M? total}$             | Spectral fit      |
| Dynamical fraction           | 40% ( $\approx 1 - \beta_i$ )               | EP-11             |
| Ejecta velocity              | $0.1c$ (blue), $0.3c$ (red)                 | Kilonova spectral |
| $Y_e$                        | $\approx 0.1$ (neutron-rich)                | EP-11 calibration |
| r-process fraction $A > 140$ | $\approx 95\%$ solar                        | EP-11             |
| $U_{b,i}$ (calculated)       | $\approx 2.3 \times 10^{-3} \text{ M?/Gyr}$ | EP-11             |

**UQFF equations:**  $U_{b,i}$  feeding outflows,  $Y_e = 1/(1 + e^{[SCm]/[UA]})$

**Verification:** EP-11 (PAPER\_109), LIGO/Virgo 2017 + 2025 NR simulations

### 5.2 AT2024tvd (Astrophysical Transient)

**Role in UQFF:** NS merger analog for triadic system update (2024).

| Parameter      | Value                   | Source           |
|----------------|-------------------------|------------------|
| Discovery      | 2024                    | ATel/ZTF         |
| Classification | NS merger analog        | arXiv 2506.04440 |
| Q_wave update  | Extends 47-system array | Grok corpus      |

**UQFF equations:**  $F_{U,\text{Triadic}}$  transient variant

**Verification:** arXiv:2506.04440, ZTF 2024

## 6. Exoplanet Systems

### 6.1 TOI 1227 b (Exoplanet)

**Role in UQFF:**  $U_{b,i}$  calibration for planetary mass loss.

| Parameter      | Value                      | Source         |
|----------------|----------------------------|----------------|
| Mass loss rate | $\approx 10^{12}$ g/s      | TESS/JWST 2025 |
| System         | Young transiting exoplanet | TESS           |
| Q_wave         | In 47-system triadic array | Grok corpus    |

**UQFF equations:**  $U_{b,i}$  (atmospheric escape),  $F_{U, \text{Triadic}}$  for mass loss dynamics  
**Verification:** TESS/JWST 2025

## 7. Nebula and Star Formation Systems

### 7.1 Pillars of Creation (Nebula)

**Role in UQFF:** Buoyancy refinement for star-forming outflows in Ug2/Ub\_i.

| Parameter | Value                                     | Source         |
|-----------|-------------------------------------------|----------------|
| Location  | M16 (Eagle Nebula)                        | —              |
| Distance  | $\approx 6,500$ ly                        | Gaia           |
| Type      | Active star formation, pillar instability | JWST 2022/2024 |
| Q_wave    | In 47-system triadic array                | Grok corpus    |

**UQFF equations:**  $U_{b,i}$  buoyancy refinements,  $U_{g2}$  bubble, alpha-conjugate clustering  
**Verification:** JWST 2025, EP-12 alpha BEC (PAPER\_107)

### 7.2 Nebular Dynamics (Generic)

**Role in UQFF:** Dust/gas stability modeling via  $U_i$ .

| Parameter          | Value                                                                       | Source    |
|--------------------|-----------------------------------------------------------------------------|-----------|
| Context            | UQFF Drawing 32<br>Nebular system                                           | Internal  |
| Stability variable | $U_i = \lambda_i \rho_{vac, [SCm]} \rho_{vac, [UA]} \omega_s \cos(\pi t_n)$ | Framework |

**UQFF equations:**  $U_i$  (universal inertia for dust/gas stability)

## 8. Nuclear and Particle Systems

### 8.1 Pb-206 (Nuclear System)

**Role in UQFF:** Nuclear binding energy ladder,  $[SSq]$  calibration.

| Parameter                  | Value                                               | Source |
|----------------------------|-----------------------------------------------------|--------|
| $n$ (level)                | 8.2 ( $\Delta n = 0.21$ )                           | EP-04  |
| $S_n$ (neutron separation) | $\approx 2 \times [SSq] \times E_8$<br>(3.5% error) | EP-04  |

| Parameter             | Value                                      | Source          |
|-----------------------|--------------------------------------------|-----------------|
| Level spacing         | $\sim 10 \text{ MeV} = 10^{-12} \text{ J}$ | ENSDF/NNDC 2025 |
| Polynomial fit degree | $n = 8$ ( $R^2 \approx 0.95$ )             | EP-04           |
| Shell levels          | 10+ levels in NNDC database                | ENSDF           |

**UQFF equations:**  $E_n = E_0 \times 10^n$  ( $n = 8$ ),  $[SSq] = 0.57$  calibration point  
**Verification:** EP-04 (PAPER\_117), ENSDF/NNDC 2025

## 8.2 $^{12}\text{C}$ Hoyle State (Nuclear BEC)

**Role in UQFF:** Bose term  $N_B$  calibration in  $U_m$ .

| Parameter               | Value                               | Source              |
|-------------------------|-------------------------------------|---------------------|
| Energy                  | 7.65 MeV                            | Hoyle 1954          |
| Configuration           | 3-alpha BEC ( $N_B = 3$ )           | Tohsaki AMD         |
| Condensate fraction     | 70-90%                              | Tohsaki et al. 2001 |
| $T_c$ (nuclear BEC)     | $\approx 1.2 \times 10^6 \text{ K}$ | EP-12 calculation   |
| LENR $\Delta T_c$ shift | $\approx 300 \text{ K}$             | EP-12 UQFF          |
| $^{16}\text{O}$ analog  | 4-alpha BEC ( $N_B = 4$ )           | Funaki et al.       |

**UQFF equations:**  $N_B$  term in  $U_m$ , LENR  $T_c$  shift via  $E_{react}$  scaling  
**Verification:** EP-12 (PAPER\_107), Tohsaki et al. arXiv:1103.3940

## 9. Compact Object Systems

### 9.1 Magnetar SGR 1745-2900

**Role in UQFF:** Full  $g_{\text{Magnetar}}$  equation anchor;  $B/B_{crit}$  calibration.

| Parameter                | Value                                   | Source        |
|--------------------------|-----------------------------------------|---------------|
| Type                     | Soft gamma repeater magnetar            | Chandra/Swift |
| $B$ field                | $\sim 10^{10}\text{-}10^{11} \text{ T}$ | Swift 2025    |
| $B_{crit}$               | $4.4 \times 10^{13} \text{ T}$          | QED critical  |
| $B/B_{crit}$             | $\sim 10^{-3}$ to $10^{-2}$             | Calibrated    |
| Mass $M_{ns}$            | $\approx 1.4 \text{ M?}$                | Standard NS   |
| Galactic center distance | $\sim 0.1 \text{ pc}$ from Sgr A*       | VLBI          |

**UQFF equations:**  $g_{\text{Magnetar}}(r, t) = (GM/r^2)(1+H(z)\cdot t)(1-B/B_{crit}) + (GM_{BH}/r_{BH}^2) + U_{g1} + U_{g2} + U_{g3} + U_{g4} + \Lambda c^2/3$   
**Verification:** Chandra/Swift 2025, PAPER\_013 (spin-down), PAPER\_121 (full equation)  
**See:** PAPER\_121 for the complete 50+ term expansion

## 10. Galactic Structural Systems

### 10.1 Galactic Disk

**Role in UQFF:** Stability analysis for  $U_{b,i}$ , galactic rotation curve.

| Parameter  | Value                           | Source                   |
|------------|---------------------------------|--------------------------|
| $\omega_g$ | $7.3 \times 10^{-16}$ rad/s     | Galactic rotation        |
| $v_{gal}$  | 220 km/s                        | Milky Way rotation curve |
| $r_{8kpc}$ | 8 kpc = $2.47 \times 10^{20}$ m | IAU 2012                 |

**UQFF equations:**  $U_{b,i}$  stability,  $U_{g4}$  BH interaction

**Verification:** Gaia DR4 2025

## 11. AGN and Cosmic Ray Systems

### 11.1 LLAGNs (Low-Luminosity AGNs)

**Role in UQFF:** CRP neutrino flux at  $\sim$ IceCube background level.

| Parameter        | Value                                                                                                | Source          |
|------------------|------------------------------------------------------------------------------------------------------|-----------------|
| Neutrino flux    | $\sim$ IceCube background $10^{-18}$ GeV <sup>1</sup> cm <sup>2</sup> s <sup>1</sup> sr <sup>1</sup> | IceCube 2022    |
| Dominant process | pp collisions < 0.1 PeV                                                                              | EP-10           |
| SED peak         | 0.05 PeV                                                                                             | EP-10 PAPER_108 |
| Spectral index   | $\Gamma \approx 2.37$                                                                                | IceCube HESE    |

**UQFF equations:** CRP term  $\sum D_E \partial^2 n / \partial p^2 \cdot e^{-\gamma t}$ , pp/p? SED

**Verification:** EP-10 (PAPER\_108), IceCube 2022

### 11.2 High-Energy Cosmic Ray Sources

**Role in UQFF:** CRP  $p_{max}$  calibration.

| Parameter      | Value                             | Source                 |
|----------------|-----------------------------------|------------------------|
| $p_{max}$      | $10^{16}$ eV                      | Fermi/Chandra 2025     |
| $n(p) \propto$ | $p^{-2.2} e^{-p/p_{max}}$         | Fokker-Planck solution |
| $D_E \propto$  | $E^{0.5}$ (Kolmogorov turbulence) | CRP physics            |

**UQFF equations:** Fokker-Planck equation, CRP term in F\_U

**Verification:** Fermi/Chandra/Parker 2025



## 12. Cosmological Systems

### 12.1 Solar Flares

**Role in UQFF:**  $U_{g1}$  dipole calibration,  $B_s = 0.4$  T upper bound.

| Parameter     | Value    | Source        |
|---------------|----------|---------------|
| $B_s$ (flare) | 0.4 T    | GOES 2025     |
| $B_s$ (quiet) | 0.0001 T | GOES baseline |

**UQFF equations:**  $U_{g1}$  dipole term

**Verification:** GOES 2025

### 12.2 Intergalactic Medium

**Role in UQFF:**  $\rho_{vac}$  ratio calibration for vacuum hierarchy.

| Parameter             | Value                                                                    | Source          |
|-----------------------|--------------------------------------------------------------------------|-----------------|
| $\rho_{vac}$ ratios   | $\sim 10^{-38}$ (low/high components)                                    | EP-08           |
| $\lambda_{vac}$ range | $\sim 10^{-9}$ J/m <sup>3</sup> (DE level)                               | JCAP 2025       |
| DM density            | $\sim 0.47$ GeV/cm <sup>3</sup> = $8.4 \times 10^{-25}$ J/m <sup>3</sup> | JCAP01(2025)021 |
| Primordial DM         | $\sim 10^{-26}$ J/m <sup>3</sup>                                         | JCAP07(2025)033 |

**UQFF equations:**  $\lambda_{vac} = \sum(f_i E_i)/V$ ,  $\rho_{vac}$  alignment

**Verification:** EP-08 (PAPER\_118), JCAP 2024/2025

### 12.3 Quasar Jets (General Class)

**Role in UQFF:** Fluid/Navier-Stokes jet modeling with  $U_{b,i}$  asymmetry.

| Parameter           | Value                       | Source             |
|---------------------|-----------------------------|--------------------|
| Jet velocity        | $\sim 0.7c$ - $0.99c$       | VLBI proper motion |
| Asymmetry mechanism | $\cos(\omega t_n)$ reversal | UQFF $U_{b,i}$     |

**UQFF equations:**  $U_{b,i}$ , Navier-Stokes in jet fluid

**Verification:** Chandra/MNRAS general

## 13. Q\_wave\_47 Statistical Summary

The 47-system Q\_wave distribution (from the UQFF verification corpus) encompasses a subset of the 24 systems above plus additional intermediate systems:

| Statistic             | Value                               |
|-----------------------|-------------------------------------|
| Mean $\bar{Q}_{wave}$ | $3.97 \times 10^4 \text{ J/m}^3$    |
| Standard deviation    | $5.11 \times 10^4 \text{ J/m}^3$    |
| Jarque-Bera           | 8.78 ( $p = 0.012$ )                |
| Leptokurtosis         | 0.037                               |
| Distribution          | Non-normal (heavy-tailed)           |
| $N$ systems           | 47 (superset of 24 catalog systems) |

The non-normal distribution confirms that UQFF  $Q_{wave}$  energy is not uniformly distributed across system types — triadic/turbulent systems (Westerlund 2, Pillars) contribute heavy tails, while individual systems (LLAGNs, nuclear) contribute near-zero floor values.

#### 14. Cross-Reference Table: Systems by Whitepaper

| System                      | Type            | Primary Paper        | EP       |
|-----------------------------|-----------------|----------------------|----------|
| Sun                         | Stellar         | PAPER_114            | EP-07    |
| Westerlund 2                | Star cluster    | $Q_{wave}$ 47 corpus | —        |
| Sgr A*                      | SMBH            | PAPER_110            | EP-06    |
| G359.13142-0.20005          | High-z          | JWST corpus          | —        |
| 3C 273                      | Quasar          | PAPER_115            | EP-09    |
| RACS J0320-35               | Quasar          | PAPER_111            | EP-01    |
| Blazars (4LAC)              | Blazars         | PAPER_113            | EP-05    |
| PLCK G287.0+32.9            | Galaxy cluster  | Triadic corpus       | —        |
| ASKAP J1832-0911            | Radio transient | Triadic corpus       | —        |
| PSZ2 G181.06+48.47          | Galaxy cluster  | Triadic corpus       | —        |
| GW170817                    | BNS merger      | PAPER_109            | EP-11    |
| AT2024tvd                   | Transient       | Triadic corpus       | —        |
| TOI 1227 b                  | Exoplanet       | Triadic corpus       | —        |
| Pillars of Creation         | Nebula          | Triadic corpus       | —        |
| Nebular Dynamics            | Generic nebula  | Framework            | —        |
| Pb-206                      | Nuclear         | PAPER_117            | EP-04    |
| $^{12}\text{C}$ Hoyle State | Nuclear BEC     | PAPER_107            | EP-12    |
| SGR 1745-2900               | Magnetar        | PAPER_013/121        | —        |
| Galactic Disk               | Galactic        | PAPER_110            | —        |
| LLAGNs                      | AGN class       | PAPER_108            | EP-10    |
| CR Sources                  | High-energy     | PAPER_088            | —        |
| Solar Flares                | Sun             | PAPER_114            | EP-07    |
| Intergalactic Medium        | Cosmological    | PAPER_118            | EP-08    |
| Quasar Jets                 | AGN class       | PAPER_111/115        | EP-01/09 |

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**Title:** Complete Catalog of Astronomical Systems Used in UQFF Calculations — Parameters, Verification Sources, and Equation Assignments

**Author:** Daniel T. Murphy

**Framework:** UQFF Star-Magic ( $\tau = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ ,  $\beta_i = 0.61$ )

**Date:** March 2026

**Source:** Grok thread 2fe4fa3e — DeepSearch extraction of “UQFF Equations Across Astrophysical Systems\_22Sept2025.docx” (393 pages)

**Index Slot:** §1.16 UQFF Equation Systems Reference

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## Abstract

The UQFF framework has been validated and applied across 24 unique astrophysical systems spanning stellar, galactic, nuclear, particle, and cosmological domains. This paper provides a complete parameter catalog for all 24 systems, organized by type, with UQFF equation assignments, calibrated parameter values, verification data sources, and cross-references to existing whitepapers. The catalog is extracted from the 393-page “UQFF Equations Across Astrophysical Systems” corpus (verified Sept 22, 2025) as the authoritative single-document reference for system parameters used in any UQFF calculation.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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## 1. Stellar and Solar Systems

### 1.1 Sun (Solar System)

**Role in UQFF:** Heliosphere modeling via Ug2 bubble, solar wind calibration.

| Parameter                         | Value                                             | Source             |
|-----------------------------------|---------------------------------------------------|--------------------|
| $M_s$                             | $1.989 \times 10^{30} \text{ kg}$                 | IAU 2015           |
| $R_b$ (heliosphere bubble radius) | $1.496 \times 10^{13} \text{ m} = 100 \text{ AU}$ | Voyager 1 crossing |

| Parameter                      | Value                                         | Source                       |
|--------------------------------|-----------------------------------------------|------------------------------|
| $\omega_s$ (rotation rate)     | $2.5 \times 10^{-6}$ rad/s                    | Solar sidereal rotation      |
| $B_s$ (surface magnetic field) | 0.0001-0.4 T                                  | GOES 2025                    |
| $\delta_{sw}$                  | $0.01 = [SSq]/57$                             | EP-07, PSP CDAWeb<br>E01-E17 |
| $v_{sw}$                       | $5 \times 10^5$ m/s                           | EP-07 PSP baseline           |
| $\rho_{sw}$                    | $\approx 8 \times 10^{-21}$ kg/m <sup>3</sup> | Parker Solar Probe           |
| $H_{SCm}$                      | $\approx 1$                                   | Heliosphere SCm factor       |
| $T_s$ (surface temperature)    | 5778 K                                        | IAU                          |

**UQFF equations:**  $U_{g2}$  (bubble),  $U_m$  (disk strings),  $U_{b,i}$  (expansion)

**Verification:** EP-07 (PAPER\_114), Parker Solar Probe CDAWeb 2025

**Q\_wave contribution:**  $Q_{wave,\odot}$  included in 47-system mean  $3.97 \times 10^4$  J/m<sup>3</sup>

## 1.2 Westerlund 2 (Star Cluster)

**Role in UQFF:** Triadic master for turbulent outflow modeling.

| Parameter  | Value                                                       | Source             |
|------------|-------------------------------------------------------------|--------------------|
| Distance   | $\approx 8$ kpc                                             | Gaia DR3           |
| Mass       | $\sim 10^4$ M?                                              | JWST 2025          |
| Age        | $\sim 2$ Myr                                                | Membership surveys |
| SFR        | High (active star formation)                                | JWST 2024          |
| $Q_{wave}$ | $\approx 3.97 \times 10^4$ J/m <sup>3</sup><br>(mean stack) | 47-system array    |

**UQFF equations:**  $F_{U,Triadic}$  with  $n = 13$  (plasma),  $U_m$  (turbulence)

**Verification:** JWST 2025 images, 47-system Q\_wave stats (Jarque-Bera = 8.78,  $p = 0.012$ )

**Cross-reference:** Q\_wave\_47 statistical system

## 2. Black Hole and Galactic Center Systems

### 2.1 Sagittarius A\* (Sgr A\*)

**Role in UQFF:** SMBH anchor for  $U_{g4}$  (BH-star interaction), galactic distance  $d_g$ .

| Parameter    | Value                                              | Uncertainty  | Source                     |
|--------------|----------------------------------------------------|--------------|----------------------------|
| $M_{bh}$     | $8.15 \times 10^{36}$ kg<br>$= 4.1 \times 10^6$ M? | $\pm 4.68\%$ | Gaia DR4 2025              |
| $d_g$        | $2.44 \times 10^{20}$ m<br>$= 25,800$ ly           | $\pm 4.51\%$ | Gaia DR4<br>(PAPER_110)    |
| $d_g$ (UQFF) | $2.55 \times 10^{20}$ m<br>$= 27,000$ ly           | —            | UQFF calibrated            |
| $\omega_g$   | $7.3 \times 10^{-16}$<br>rad/s                     | —            | Galactic rotation<br>curve |

| Parameter | Value                                        | Uncertainty | Source                  |
|-----------|----------------------------------------------|-------------|-------------------------|
| $U_{g4}$  | $1.8937 \times 10^{-23}$<br>J/m <sup>3</sup> | —           | PAPER_110<br>calculated |

**UQFF equations:**  $U_{g4}$ ,  $E_{react}$  (from BH-scale reactivity)

**Verification:** EP-06 (PAPER\_110), Gaia DR3/DR4 2025

**Note:** The UQFF uses  $d_g = 2.55 \times 10^{20}$  m (Galactic longitude model) vs Gaia's  $2.44 \times 10^{20}$  m (4.5% difference within observational uncertainty).

## 2.2 G359.13142-0.20005

**Role in UQFF:** High-z object for shear map analysis and triadic analogs.

| Parameter            | Value                                  | Source         |
|----------------------|----------------------------------------|----------------|
| Galactic coordinates | $l = 359.13^\circ$ , $b = -0.20^\circ$ | NISP survey    |
| $\delta_\tau$        | $\approx 0.05$                         | JWST NISP 2025 |

**UQFF equations:**  $F_{U, \text{Triadic}}$ , shear maps,  $\delta_\tau$  calibration

**Verification:** JWST 2025 data, NISP catalog

## 3. Quasar and Blazar Systems

### 3.1 3C 273 (Quasar)

**Role in UQFF:** Asymmetric jet verification of  $t_n$  reversals.

| Parameter            | Value                                    | Source                |
|----------------------|------------------------------------------|-----------------------|
| Redshift $z$         | 0.158                                    | Schmidt 1963          |
| Luminosity $L$       | $\sim 10^{46}$ erg/s = $10^{39}$ W       | MNRAS 2025            |
| Jet length           | $\sim 200$ kpc (apparent), 65 kpc proper | VLBI                  |
| Brightness asymmetry | $> 100 : 1$                              | MNRAS 482, 743 (2019) |
| $N$ (reversal count) | 13 sequential $t_n$ reversals            | PAPER_115             |
| Asymmetry ratio $R$  | $130 > 100$ target                       | EP-09 calculation     |

**UQFF equations:**  $U_{b,i}$  with  $\cos(\pi t_n)$ ,  $U_m$ ; ratio  $R = |\cos(\pi t_{n1})/\cos(\pi t_{n2})|^N$

**Verification:** EP-09 (PAPER\_115), MNRAS 482/743 2019, arXiv:2412.18250

### 3.2 RACS J0320-35 (Quasar)

**Role in UQFF:** Navier-Stokes jet fluid verification of  $U_{b,i}$  asymmetry.

| Parameter                   | Value               | Source            |
|-----------------------------|---------------------|-------------------|
| Jet asymmetry ratio $R$     | $\approx 1.5$       | Chandra 2025      |
| $\tau_{dissip}$             | 9 Gyr               | EP-01 calculation |
| $\cos(\omega t_n)$ reversal | Sign flip confirmed | EP-01             |

| Parameter | Value | Source |
|-----------|-------|--------|
|-----------|-------|--------|

**UQFF equations:**  $U_{b,i}$ , Navier-Stokes fluid in jets

**Verification:** EP-01 (PAPER\_111), Chandra X-ray Observatory 2025

### 3.3 Blazars (Fermi LAT 4LAC)

**Role in UQFF:**  $E_{react}$  decay rate  $\kappa$  calibration across 3,743 sources.

| Parameter            | Value                                             | Source          |
|----------------------|---------------------------------------------------|-----------------|
| Sample size          | 3,743 sources (4LAC-DR3)                          | Fermi HEASARC   |
| Luminosity range     | $10^{39}$ - $10^{47}$ W                           | 4LAC-DR3        |
| $\kappa$ (measured)  | $0.000497 \pm 5\%$ day <sup>-1</sup>              | EP-05 PAPER_113 |
| 8-bin $\kappa$ error | All $< 5\%$                                       | EP-05           |
| $E_{react}$ range    | $10^{39}$ - $10^{47}$ W/m <sup>3</sup> (8 L bins) | EP-05           |

**UQFF equations:**  $E_{react} = 10^{46} e^{-\kappa t}$ ,  $L \propto E_{react}$  by blazar class

**Verification:** EP-05 (PAPER\_113), Fermi LAT 4LAC-DR4, HEASARC 2025

## 4. Galaxy Cluster and Radio Systems

### 4.1 PLCK G287.0+32.9 (Galaxy Cluster)

**Role in UQFF:** Double radio relic calibration for triadic systems.

| Parameter   | Value                            | Source      |
|-------------|----------------------------------|-------------|
| Type        | Galaxy cluster (double relic)    | Planck PSZ2 |
| $M_{500,X}$ | Calibrated to triadic mass scale | PSZ2        |
| Q_wave      | In 47-system array               | Grok corpus |

**UQFF equations:**  $F_{U,Triadic}$ ,  $M_{500,X}$  calibration

**Verification:** Planck/PSZ2 catalog

### 4.2 ASKAP J1832-0911 (Radio Transient)

**Role in UQFF:** Transient source for Q\_wave array updates and triadic analogs.

| Parameter      | Value              | Source      |
|----------------|--------------------|-------------|
| Survey         | ASKAP 2025         | ASKAP       |
| Classification | Radio transient    | ASKAP       |
| Q_wave         | In 47-system array | Grok corpus |

**UQFF equations:**  $F_{U,Triadic}$  transient variant

**Verification:** ASKAP 2025 survey

### 4.3 PSZ2 G181.06+48.47 (Galaxy Cluster)

**Role in UQFF:** Low-mass merger for double relic analogs.

| Parameter   | Value                            | Source      |
|-------------|----------------------------------|-------------|
| $M_{500,X}$ | $2.57 \times 10^{14} \text{ M?}$ | Planck 2025 |
| Merger type | Low-mass merger                  | Planck      |
| Q_wave      | In 47-system array               | Grok corpus |

**UQFF equations:**  $F_{U,\text{Triadic}}$ , merger relic analog

**Verification:** Planck 2025, PSZ2 catalog

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## 5. Transient and Merger Systems

### 5.1 GW170817 (BNS Merger)

**Role in UQFF:**  $\beta_i$  calibration via r-process ejecta fraction.

| Parameter                    | Value                                       | Source            |
|------------------------------|---------------------------------------------|-------------------|
| Event date                   | August 17, 2017                             | LIGO/Virgo        |
| Total merger mass            | $\approx 2.8 \text{ M?}$                    | LIGO 2017         |
| Ejecta mass $M_{ej}$         | $\approx 0.05 \text{ M? total}$             | Spectral fit      |
| Dynamical fraction           | 40% ( $\approx 1 - \beta_i$ )               | EP-11             |
| Ejecta velocity              | 0.1c (blue), 0.3c (red)                     | Kilonova spectral |
| $Y_e$                        | $\approx 0.1$ (neutron-rich)                | EP-11 calibration |
| r-process fraction $A > 140$ | $\approx 95\%$ solar                        | EP-11             |
| $U_{b,i}$ (calculated)       | $\approx 2.3 \times 10^{-3} \text{ M?/Gyr}$ | EP-11             |

**UQFF equations:**  $U_{b,i}$  feeding outflows,  $Y_e = 1/(1 + e^{[SCm]/[UA]})$

**Verification:** EP-11 (PAPER\_109), LIGO/Virgo 2017 + 2025 NR simulations

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### 5.2 AT2024tvd (Astrophysical Transient)

**Role in UQFF:** NS merger analog for triadic system update (2024).

| Parameter      | Value                   | Source           |
|----------------|-------------------------|------------------|
| Discovery      | 2024                    | ATel/ZTF         |
| Classification | NS merger analog        | arXiv 2506.04440 |
| Q_wave update  | Extends 47-system array | Grok corpus      |

**UQFF equations:**  $F_{U,\text{Triadic}}$  transient variant

**Verification:** arXiv:2506.04440, ZTF 2024

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## 6. Exoplanet Systems

### 6.1 TOI 1227 b (Exoplanet)

**Role in UQFF:**  $U_{b,i}$  calibration for planetary mass loss.

| Parameter      | Value                      | Source         |
|----------------|----------------------------|----------------|
| Mass loss rate | $\approx 10^{12}$ g/s      | TESS/JWST 2025 |
| System         | Young transiting exoplanet | TESS           |
| Q_wave         | In 47-system triadic array | Grok corpus    |

**UQFF equations:**  $U_{b,i}$  (atmospheric escape),  $F_{U, \text{Triadic}}$  for mass loss dynamics

**Verification:** TESS/JWST 2025

## 7. Nebula and Star Formation Systems

### 7.1 Pillars of Creation (Nebula)

**Role in UQFF:** Buoyancy refinement for star-forming outflows in Ug2/Ub\_i.

| Parameter | Value                                     | Source         |
|-----------|-------------------------------------------|----------------|
| Location  | M16 (Eagle Nebula)                        | —              |
| Distance  | $\approx 6,500$ ly                        | Gaia           |
| Type      | Active star formation, pillar instability | JWST 2022/2024 |
| Q_wave    | In 47-system triadic array                | Grok corpus    |

**UQFF equations:**  $U_{b,i}$  buoyancy refinements,  $U_{g2}$  bubble, alpha-conjugate clustering

**Verification:** JWST 2025, EP-12 alpha BEC (PAPER\_107)

### 7.2 Nebular Dynamics (Generic)

**Role in UQFF:** Dust/gas stability modeling via  $U_i$ .

| Parameter          | Value                                                                       | Source    |
|--------------------|-----------------------------------------------------------------------------|-----------|
| Context            | UQFF Drawing 32<br>Nebular system                                           | Internal  |
| Stability variable | $U_i = \lambda_i \rho_{vac, [SCm]} \rho_{vac, [UA]} \omega_s \cos(\pi t_n)$ | Framework |

**UQFF equations:**  $U_i$  (universal inertia for dust/gas stability)

## 8. Nuclear and Particle Systems

### 8.1 Pb-206 (Nuclear System)

**Role in UQFF:** Nuclear binding energy ladder,  $[SSq]$  calibration.



| Parameter                  | Value                                               | Source          |
|----------------------------|-----------------------------------------------------|-----------------|
| $n$ (level)                | 8.2 ( $\Delta n = 0.21$ )                           | EP-04           |
| $S_n$ (neutron separation) | $\approx 2 \times [SSq] \times E_8$<br>(3.5% error) | EP-04           |
| Level spacing              | $\sim 10 \text{ MeV} = 10^{-12} \text{ J}$          | ENSDF/NNDC 2025 |
| Polynomial fit degree      | $n = 8$ ( $R^2 \approx 0.95$ )                      | EP-04           |
| Shell levels               | 10+ levels in NNDC database                         | ENSDF           |

**UQFF equations:**  $E_n = E_0 \times 10^n$  ( $n = 8$ ),  $[SSq] = 0.57$  calibration point

**Verification:** EP-04 (PAPER\_117), ENSDF/NNDC 2025

## 8.2 $^{12}\text{C}$ Hoyle State (Nuclear BEC)

**Role in UQFF:** Bose term  $N_B$  calibration in  $U_m$ .

| Parameter               | Value                               | Source              |
|-------------------------|-------------------------------------|---------------------|
| Energy                  | 7.65 MeV                            | Hoyle 1954          |
| Configuration           | 3-alpha BEC ( $N_B = 3$ )           | Tohsaki AMD         |
| Condensate fraction     | 70-90%                              | Tohsaki et al. 2001 |
| $T_c$ (nuclear BEC)     | $\approx 1.2 \times 10^6 \text{ K}$ | EP-12 calculation   |
| LENR $\Delta T_c$ shift | $\approx 300 \text{ K}$             | EP-12 UQFF          |
| $^{16}\text{O}$ analog  | 4-alpha BEC ( $N_B = 4$ )           | Funaki et al.       |

**UQFF equations:**  $N_B$  term in  $U_m$ , LENR  $T_c$  shift via  $E_{react}$  scaling

**Verification:** EP-12 (PAPER\_107), Tohsaki et al. arXiv:1103.3940

## 9. Compact Object Systems

### 9.1 Magnetar SGR 1745-2900

**Role in UQFF:** Full  $g_{\text{Magnetar}}$  equation anchor;  $B/B_{crit}$  calibration.

| Parameter                | Value                                   | Source        |
|--------------------------|-----------------------------------------|---------------|
| Type                     | Soft gamma repeater magnetar            | Chandra/Swift |
| $B$ field                | $\sim 10^{10}\text{-}10^{11} \text{ T}$ | Swift 2025    |
| $B_{crit}$               | $4.4 \times 10^{13} \text{ T}$          | QED critical  |
| $B/B_{crit}$             | $\sim 10^{-3}$ to $10^{-2}$             | Calibrated    |
| Mass $M_{ns}$            | $\approx 1.4 \text{ M?}$                | Standard NS   |
| Galactic center distance | $\sim 0.1 \text{ pc}$ from Sgr A*       | VLBI          |

**UQFF equations:**  $g_{\text{Magnetar}}(r, t) = (GM/r^2)(1+H(z)\cdot t)(1-B/B_{crit}) + (GM_{BH}/r_{BH}^2) + U_{g1} + U_{g2} + U_{g3} + U_{g4} + \Lambda c^2/3$

**Verification:** Chandra/Swift 2025, PAPER\_013 (spin-down), PAPER\_121 (full equation)

**See:** PAPER\_121 for the complete 50+ term expansion

## 10. Galactic Structural Systems

### 10.1 Galactic Disk

**Role in UQFF:** Stability analysis for  $U_{b,i}$ , galactic rotation curve.

| Parameter  | Value                           | Source                   |
|------------|---------------------------------|--------------------------|
| $\omega_g$ | $7.3 \times 10^{-16}$ rad/s     | Galactic rotation        |
| $v_{gal}$  | 220 km/s                        | Milky Way rotation curve |
| $r_{8kpc}$ | 8 kpc = $2.47 \times 10^{20}$ m | IAU 2012                 |

**UQFF equations:**  $U_{b,i}$  stability,  $U_{g4}$  BH interaction

**Verification:** Gaia DR4 2025

## 11. AGN and Cosmic Ray Systems

### 11.1 LLAGNs (Low-Luminosity AGNs)

**Role in UQFF:** CRP neutrino flux at  $\sim$ IceCube background level.

| Parameter        | Value                                                                                                | Source          |
|------------------|------------------------------------------------------------------------------------------------------|-----------------|
| Neutrino flux    | $\sim$ IceCube background $10^{-18}$ GeV <sup>1</sup> cm <sup>2</sup> s <sup>1</sup> sr <sup>1</sup> | IceCube 2022    |
| Dominant process | pp collisions < 0.1 PeV                                                                              | EP-10           |
| SED peak         | 0.05 PeV                                                                                             | EP-10 PAPER_108 |
| Spectral index   | $\Gamma \approx 2.37$                                                                                | IceCube HESE    |

**UQFF equations:** CRP term  $\sum D_E \partial^2 n / \partial p^2 \cdot e^{-\gamma t}$ , pp/p? SED

**Verification:** EP-10 (PAPER\_108), IceCube 2022

### 11.2 High-Energy Cosmic Ray Sources

**Role in UQFF:** CRP  $p_{max}$  calibration.

| Parameter      | Value                             | Source                 |
|----------------|-----------------------------------|------------------------|
| $p_{max}$      | $10^{16}$ eV                      | Fermi/Chandra 2025     |
| $n(p) \propto$ | $p^{-2.2} e^{-p/p_{max}}$         | Fokker-Planck solution |
| $D_E \propto$  | $E^{0.5}$ (Kolmogorov turbulence) | CRP physics            |

**UQFF equations:** Fokker-Planck equation, CRP term in F\_U

**Verification:** Fermi/Chandra/Parker 2025

## 12. Cosmological Systems

### 12.1 Solar Flares

**Role in UQFF:**  $U_{g1}$  dipole calibration,  $B_s = 0.4$  T upper bound.

| Parameter     | Value    | Source        |
|---------------|----------|---------------|
| $B_s$ (flare) | 0.4 T    | GOES 2025     |
| $B_s$ (quiet) | 0.0001 T | GOES baseline |

**UQFF equations:**  $U_{g1}$  dipole term

**Verification:** GOES 2025

### 12.2 Intergalactic Medium

**Role in UQFF:**  $\rho_{vac}$  ratio calibration for vacuum hierarchy.

| Parameter             | Value                                                                    | Source          |
|-----------------------|--------------------------------------------------------------------------|-----------------|
| $\rho_{vac}$ ratios   | $\sim 10^{-38}$ (low/high components)                                    | EP-08           |
| $\lambda_{vac}$ range | $\sim 10^{-9}$ J/m <sup>3</sup> (DE level)                               | JCAP 2025       |
| DM density            | $\sim 0.47$ GeV/cm <sup>3</sup> = $8.4 \times 10^{-25}$ J/m <sup>3</sup> | JCAP01(2025)021 |
| Primordial DM         | $\sim 10^{-26}$ J/m <sup>3</sup>                                         | JCAP07(2025)033 |

**UQFF equations:**  $\lambda_{vac} = \sum(f_i E_i)/V$ ,  $\rho_{vac}$  alignment

**Verification:** EP-08 (PAPER\_118), JCAP 2024/2025

### 12.3 Quasar Jets (General Class)

**Role in UQFF:** Fluid/Navier-Stokes jet modeling with  $U_{b,i}$  asymmetry.

| Parameter           | Value                       | Source             |
|---------------------|-----------------------------|--------------------|
| Jet velocity        | $\sim 0.7c$ - $0.99c$       | VLBI proper motion |
| Asymmetry mechanism | $\cos(\omega t_n)$ reversal | UQFF $U_{b,i}$     |

**UQFF equations:**  $U_{b,i}$ , Navier-Stokes in jet fluid

**Verification:** Chandra/MNRAS general

## 13. Q\_wave\_47 Statistical Summary

The 47-system Q\_wave distribution (from the UQFF verification corpus) encompasses a subset of the 24 systems above plus additional intermediate systems:

| Statistic             | Value                               |
|-----------------------|-------------------------------------|
| Mean $\bar{Q}_{wave}$ | $3.97 \times 10^4 \text{ J/m}^3$    |
| Standard deviation    | $5.11 \times 10^4 \text{ J/m}^3$    |
| Jarque-Bera           | 8.78 ( $p = 0.012$ )                |
| Leptokurtosis         | 0.037                               |
| Distribution          | Non-normal (heavy-tailed)           |
| $N$ systems           | 47 (superset of 24 catalog systems) |

The non-normal distribution confirms that UQFF  $Q_{wave}$  energy is not uniformly distributed across system types — triadic/turbulent systems (Westerlund 2, Pillars) contribute heavy tails, while individual systems (LLAGNs, nuclear) contribute near-zero floor values.

#### 14. Cross-Reference Table: Systems by Whitepaper

| System                      | Type            | Primary Paper        | EP       |
|-----------------------------|-----------------|----------------------|----------|
| Sun                         | Stellar         | PAPER_114            | EP-07    |
| Westerlund 2                | Star cluster    | $Q_{wave}$ 47 corpus | —        |
| Sgr A*                      | SMBH            | PAPER_110            | EP-06    |
| G359.13142-0.20005          | High-z          | JWST corpus          | —        |
| 3C 273                      | Quasar          | PAPER_115            | EP-09    |
| RACS J0320-35               | Quasar          | PAPER_111            | EP-01    |
| Blazars (4LAC)              | Blazars         | PAPER_113            | EP-05    |
| PLCK G287.0+32.9            | Galaxy cluster  | Triadic corpus       | —        |
| ASKAP J1832-0911            | Radio transient | Triadic corpus       | —        |
| PSZ2 G181.06+48.47          | Galaxy cluster  | Triadic corpus       | —        |
| GW170817                    | BNS merger      | PAPER_109            | EP-11    |
| AT2024tvd                   | Transient       | Triadic corpus       | —        |
| TOI 1227 b                  | Exoplanet       | Triadic corpus       | —        |
| Pillars of Creation         | Nebula          | Triadic corpus       | —        |
| Nebular Dynamics            | Generic nebula  | Framework            | —        |
| Pb-206                      | Nuclear         | PAPER_117            | EP-04    |
| $^{12}\text{C}$ Hoyle State | Nuclear BEC     | PAPER_107            | EP-12    |
| SGR 1745-2900               | Magnetar        | PAPER_013/121        | —        |
| Galactic Disk               | Galactic        | PAPER_110            | —        |
| LLAGNs                      | AGN class       | PAPER_108            | EP-10    |
| CR Sources                  | High-energy     | PAPER_088            | —        |
| Solar Flares                | Sun             | PAPER_114            | EP-07    |
| Intergalactic Medium        | Cosmological    | PAPER_118            | EP-08    |
| Quasar Jets                 | AGN class       | PAPER_111/115        | EP-01/09 |

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